DEEPAK NEUPANE

SOFTWARE ENGINEER · DEVOPS · TEST AUTOMATION
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Software engineer with six years building internal applications, CI/CD infrastructure, and test automation for safety-critical systems. Acting technical lead for the team — owning priorities, escalations, and delivery decisions in the lead's absence. Track record of evaluating and rolling out tooling adopted org-wide, mentoring engineers onto frameworks I built, and serving as the primary technical contact for developers across build, pipeline, and automation issues.

EXPERIENCE

Software Engineer III, HL Mechatronics 2020 – Present

- **Software Development —** Built internal applications and automation tooling in Python, Java, C#, C, and Groovy. Developed web applications and REST services with Django over SQLite and PostgreSQL, using Redis for caching and session management. Owned the full lifecycle from requirements through containerized deployment on self-managed infrastructure.
- **DevOps & Infrastructure —** Managed CI/CD pipelines across GitLab, Bitbucket, Jenkins, and PTC Integrity. Containerized applications with Docker and deployed to Kubernetes, instrumenting services with Prometheus and Grafana for metrics and alerting. Defined repository structure and branching strategy, administered build servers, and coordinated release deliverables across teams. Primary escalation point for build and pipeline failures, unblocking developers and minimizing downtime from tooling issues.
- **Test Automation & Quality Engineering —** Led the test automation program, building Python and Shell/Batch frameworks that replaced manual regression cycles. Instrumented fault injection code to validate error handling and system resilience under failure conditions. Onboarded and mentored engineers on the framework, establishing the patterns and review practices the team adopted.
- **Technical Evaluation & Tool Selection —** Evaluated and recommended development, static analysis, and diagnostic tooling adopted across the organization, including MATLAB/Simulink, Polyspace, PTC Integrity, and the Vector toolchain.
- **Systems & Security Engineering —** Contributed to bootloader, Hardware Security Module, and application software on resource-constrained targets. Implemented and verified cybersecurity features including CMAC verification. Performed penetration testing, fault injection testing, and static code analysis.
- **Cross-Functional Support —** Bench tested diagnostic and calibration services (UDS, XCP) with factory engineers to resolve production line issues under time pressure.

TECHNICAL SKILLS

Languages Python, Java, C#, C, Groovy

Frameworks Django, React

DevOps & CI/CD Jenkins, GitLab, Bitbucket, Git, PTC Integrity

Databases & Caching SQLite, PostgreSQL, Redis

Infrastructure & Monitoring Docker, Kubernetes, Prometheus, Grafana

Test & Quality Python, Shell/Batch, ECUTest, fault injection testing, penetration testing, static code analysis (Polyspace)

Systems & Embedded Tasking TriCore and Wind River compilers, iSYSTEM (winIDEA, BlueBox), MATLAB/Simulink

Domain Tooling Vector CANoe, CANape, DaVinci Configurator; UDS, XCP

Leadership Acting technical lead, mentoring & onboarding, developer enablement, technical evaluation and tool selection

EDUCATION & CERTIFICATIONS

M.S., Software Engineering and Information Technology 2024

Oakland University — Rochester, Michigan

Google Cloud Associate Cloud Engineer In progress